



The circuit's working is simple.

Construction and testing

An actual-size, single-side PCB layout for the smoke alarm is shown in Fig. 4, and its component layout is shown in Fig. 5. After assembling the circuit on the PCB, enclose it in a suitable box. Fix LED1 and CON2 on the front of the box. Connect the sensor's three pins to CON2 using three different colour flexible wires. Position the MQ2 sensor module such that it is exposed to maximum smoke during a mishap. After assembling the circuit, install the unit at a suitable place, ensuring easy access by the sensor in the event of smoke production or excessive smoke in the area.

After properly installing the device at the venue to be protected, connect the sensor to the PCB of the assembled circuit. Now, your smoke alarm is ready to use.

EFY note. This circuit uses a 3-pin MQ2 sensor module, but you can use a 4-pin MQ2 sensor module too. In that case, use output pin A0 and avoid pin D0.

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-make-smoke-alarm-using-um3562> 

S.C. Dwivedi is an electronics enthusiast and circuit designer at EFY